

MIT OpenCourseWare

6.006

Introduction to Algorithms 6.006 — Introduction to Algorithms May 2026

EXTREMELY HARD – MIT LEVEL

Time Limit: 3 hours**Total Marks: 100**

1. (4 points) Prove the Adleman theorem: $\text{BPP} \subseteq \text{P/poly}$. Show that any randomized polynomial-time algorithm can be derandomized with polynomial-size advice.
2. (2 points) Prove the VC-dimension bound for PAC learning: if a hypothesis class $\mathcal{H}$ has VC-dimension d , then the sample complexity for (ε, δ) -PAC learning is $m \geq \Omega(\frac{1}{\varepsilon}(d + \log(1/\delta)))$. Provide the upper bound as well.
3. (4 points) Prove that EXP^{NP} contains languages that require exponential-size circuits (Karp-Lipton theorem variant). Show that if $\text{NP} \subseteq \text{P/poly}$ then $\text{PH} = \Sigma_2^P$.
4. (3 points) Prove the Byzantine Generals Problem has no solution for $n \leq 3m$ generals with m traitors in the oral messages model. Show the $n \geq 3m + 1$ necessary condition.
5. (3 points) Prove that the minimum distance of a linear code can be computed using the parity check matrix and show that the decisional version is NP-complete (Berlekamp-Welch theorem variant).
6. (4 points) Prove the CAP theorem: a distributed data store cannot simultaneously provide more than two of Consistency, Availability, and Partition tolerance. Formalize the theorem using a proof by contradiction with atomic registers.
7. (3 points) Prove the Chernoff bound: for independent Bernoulli random variables $X_1, \dots, X_n$ with $\Pr[X_i = 1] = p_i$, let $X = \sum X_i$ and $\mu = \mathbb{E}[X]$. Show that $\Pr[X \geq (1 + \delta)\mu] \leq (\frac{e^\delta}{(1+\delta)^{1+\delta}})^\mu$ for all $\delta > 0$.
8. (7 points) Prove that graph isomorphism is in $\text{NP} \cap \text{coAM}$. Show that GI is unlikely to be NP-complete (if it were, PH collapses).
9. (5 points) Prove that the matrix multiplication exponent ω satisfies $2 \leq \omega < 2.373$ by analyzing the Coppersmith-Winograd algorithm and its subsequent improvements.
10. (5 points) Prove the parallel repetition theorem for interactive proofs: if a verifier accepts with probability p in a single round, then the verifier accepts in n parallel repetitions with probability at most $p^{\Omega(n)}$ (for certain protocols).

11. (5 points) Prove that the amortized cost of a sequence of n operations in a Fibonacci heap is $O(\log n)$ for extract-min and $O(1)$ for all other operations, using the potential function method.
12. (7 points) Prove the Cook-Levin theorem: SAT is NP-complete. Construct a polynomial-time reduction from any NP language L to SAT.
13. (2 points) Prove an $\Omega(n \log n)$ lower bound for sorting in the comparison-based model using the adversary method. Then apply the adversary argument to prove a lower bound for finding the k -th smallest element.
14. (4 points) Prove the PCP theorem: $\text{NP} = \text{PCP}(\log n, 1)$. Construct a verifier for SAT that reads $O(1)$ bits of the proof and uses $O(\log n)$ random bits.
15. (4 points) Prove that parity cannot be computed by a polynomial-size constant-depth circuit (AC^0) using the switching lemma. Show that any AC^0 circuit computing parity must have size $\exp(\Omega(n^{1/(d-1)}))$.
16. (3 points) Prove correctness of the Cheney garbage collection algorithm using a formal argument about the copy phase preserving the reachability graph.
17. (6 points) Prove the space hierarchy theorem: for space-constructible $f(n) = \Omega(\log n)$, $\text{SPACE}[f(n)] \subsetneq \text{SPACE}[f(n) \log f(n)]$.
18. (4 points) Prove the Karger-Stein randomized min-cut algorithm runs in $O(n^2 \log^3 n)$ time and succeeds with high probability. Analyze the probability that a particular min-cut survives the contraction process.
19. (4 points) Prove the time hierarchy theorem: for time-constructible functions $f(n)$, there exists a language decidable in $O(f(n))$ time but not in $o(f(n)/\log f(n))$ time.
20. (3 points) Prove that directed Steiner tree is NP-hard by reducing from set cover. Then show it is approximable within $O(\log^2 n)$.
21. (4 points) Prove type soundness for the simply typed lambda calculus using the progress and preservation lemmas. Extend the proof to System F with parametric polymorphism.
22. (5 points) Prove that 3-SAT reduces to Hamiltonian Cycle in polynomial time. Construct the explicit gadget for each clause and variable, showing that a satisfying assignment corresponds to a Hamiltonian cycle in the constructed graph.
23. (3 points) Prove that problem $\text{RL} \subseteq \text{SC}$ (randomized logspace is in Steve's class) using the Nisan generator and showing that RL can be derandomized with polynomial seed length.

24. (4 points) Prove that if $P = NP$, then $BPP = P$. Conversely, show that $BPP \subseteq EXP$ and discuss the relationship between P and BPP .
25. (2 points) State and prove Rice's theorem: every nontrivial semantic property of partial recursive functions is undecidable. Show that 'program P halts on input 0' is such a property.

Solutions

Solution 1:

Let $L \in \text{BPP}$ with TM M using $m(n)$ random bits. For each input size n , the set of bad random strings for input x has size $\leq 2^{m(n)}/3$. By union bound over 2^n inputs, the total bad set has size $\leq 2^n \cdot 2^{m(n)}/3 < 2^{m(n)}$ for n large enough. So there exists a good random string r_n that works for all inputs of length n . Provide r_n as advice, giving $\text{BPP} \subseteq \text{P/poly}$.

Solution 2:

Lower bound: construct d shattered points. The adversary picks a target concept uniformly from 2^d dichotomies. To learn, we must see each point with label to distinguish. By the Sauer-Shelah lemma, $|\mathcal{H}(m)| \leq (em/d)^d$. Setting sample size $m = O(\frac{1}{\epsilon}(d \log(1/\epsilon) + \log(1/\delta)))$ gives the upper bound via the ERM principle.

Solution 3:

If $\text{NP} \subseteq \text{P/poly}$, then SAT has polynomial-size circuits. By the Karp-Lipton theorem, $\Pi_2^P \subseteq \Sigma_2^P$, collapsing PH to Σ_2^P . The proof: for any Π_2^P language, the existential player guesses the circuit for SAT and uses it to answer the universal quantifier. This shows $\text{PH} \subseteq \Sigma_2^P$, hence $\text{PH} = \Sigma_2^P$.

Solution 4:

For $m = 1, n = 3$: assume generals A (commander), B, C with B loyal, C traitor. Commander sends 'attack'. C tells B 'retreat'. B receives conflicting messages. For $n = 3m$, reduce by grouping: simulate each general as $m + 1$ generals, one traitor in each group to create inconsistency. The $n \geq 3m + 1$ condition is sufficient via the Oral Message algorithm OM(m) recursively sending $m + 1$ rounds.

Solution 5:

For a linear $[n, k]$ code with parity check matrix H , the minimum distance $d = \min_{c \neq 0, Hc^T = 0} w(c)$. Decisional problem (given G and w , is $d \leq w$?) is NP-complete via reduction from 3SAT, by constructing a code whose parity checks encode clause satisfaction. For a linear code over $\mathbb{F}_q$, finding minimum weight codeword is NP-hard.

Solution 6:

Assume all three hold. During a partition, two nodes cannot communicate. A write to node A must be visible on a read from node B (consistency). But node B must respond (availability) without contacting A (partition). So B has either stale data (no consistency) or must wait (no availability), contradiction. This shows at most two properties can hold simultaneously.

Solution 7:

Use the moment generating function: $\Pr[X \geq t] \leq \min_{\lambda > 0} e^{-\lambda t} \mathbb{E}[e^{\lambda X}] = \min_{\lambda > 0} e^{-\lambda t} \prod \mathbb{E}[e^{\lambda X_i}]$. Since $\mathbb{E}[e^{\lambda X_i}] = 1 + p_i(e^\lambda - 1) \leq \exp(p_i(e^\lambda - 1))$, we get $\Pr[X \geq (1 + \delta)\mu] \leq (\frac{e^\delta}{(1 + \delta)^{1 + \delta}})^\mu$ by optimizing $\lambda = \ln(1 + \delta)$.

Solution 8:

$\text{GI} \in \text{NP}$: a permutation is a polynomial-time verifiable certificate. $\text{GI} \in \text{coAM}$: the

verifier sends a random permutation of one graph; the prover must identify which graph. The interactive protocol has soundness $1/2$. If GI were NP -complete, then $\text{PH} = \Sigma_2^P$ (Boole'an-Karp'n theorem) because $\text{coNP} \subseteq \text{AM}$ and AM collapses if $\text{coNP} \subseteq \text{AM}$.

Solution 9:

Strassen's $O(n^{2.807})$ algorithm uses the identity $M_2(2) = 7$ giving $\omega = \log_2 7$. Coppersmith-Winograd uses the laser method and bilinear algebra to achieve $\omega < 2.376$. The key idea: construct a family of tensors T_n with border rank r_n and combining them recursively. The asymptotic inequality $\omega < \log_{n^{2/3}}(r_n)$ yields improved bounds. The current best $\omega < 2.37286$ uses refined laser method analysis.

Solution 10:

For XOR games, Raz's parallel repetition theorem shows the value decays exponentially: $\omega(G^{\otimes n}) \leq \omega(G)^{cn}$ where c depends on the game. The proof uses information theory: the winning probability for each coordinate is correlated only through the initial shared randomness, and the protocol cannot coordinate between coordinates without increasing communication.

Solution 11:

Define $\Phi = \text{number of trees} + 2 \cdot \text{number of marked nodes}$. Insert: add one tree, $\Delta\Phi = 1$, actual cost $O(1)$, amortized $O(1)$. Decrease-key: cut and cascade, each cut adds a tree but may clear marks. Extract-min: $O(\text{trees} + \text{degree})$ actual cost. By the rank bound $\text{degree} = O(\log n)$, and trees decrease by at most $O(\log n)$, giving amortized $O(\log n)$.

Solution 12:

Let $L \in \text{NP}$ via NTM M running in time $p(n)$. Construct a boolean formula ϕ with variables encoding the computation tableau of M on input x of size n . ϕ has clauses for: (1) initial configuration, (2) valid transitions using $O(p(n)^3)$ variables, (3) accepting state. ϕ is satisfiable iff M accepts x . The reduction runs in polynomial time, proving SAT is NP-hard.

Solution 13:

Any comparison tree that sorts n elements must have at least $n!$ leaves, so depth $\geq \log_2(n!) = \Omega(n \log n)$. Adversary argument: maintain a set of possible permutations consistent with comparisons so far. Each comparison eliminates at most half the remaining permutations, so $\Omega(\log(n!))$ comparisons needed. For selection, the adversary maintains a partial order ensuring at least $n - 1$ comparisons are needed for the minimum.

Solution 14:

$(\supseteq)$ is trivial. $(\subseteq)$: Build a PCP verifier for SAT. Start from the NP-complete constraint satisfaction problem with constant degree (via expander graphs). Apply the gap amplification step (repeatedly replace constraints by their AND using composition). The final verifier reads $O(1)$ proof bits. The theorem shows that approximate solutions to NP problems remain hard.

Solution 15:

Håstad's switching lemma: a random restriction of a DNF formula leaves a decision

tree of depth $O(\log s)$ with high probability. Apply this to each layer of an AC^0 circuit of depth d and size S . After $d - 1$ random restrictions, the circuit becomes a decision tree of depth $O(\log S)^{d-1}$. Since parity has decision tree depth n , $O(\log S)^{d-1} \geq n$, so $S \geq 2^{\Omega(n^{1/(d-1)})}$.

Solution 16:

Cheney's algorithm uses a semi-space collector. Maintain *fromspace* and *tospace*. The scan pointer iterates through *tospace* while the alloc pointer advances. Invariant: all reachable objects from roots have been copied or are in the queue. Each edge traversed preserves connectivity: $\forall x$ reachable from root, x is either in *tospace* (scanned or unscanned) or reachable from unscanned objects. Termination: when scan = free, all reachable objects are in *tospace*.

Solution 17:

Diagonalization: construct a language L decidable in $O(f(n) \log f(n))$ space but not in $o(f(n))$ space. The diagonalizing machine D on input x simulates M_x on x using at most $f(|x|)$ space and if M_x halts within $2^{O(f)}$ steps, outputs the opposite. The simulation uses a counter of $O(\log f)$ bits. The $\log f(n)$ factor is needed for configuration counting to detect infinite loops.

Solution 18:

Contract edges uniformly until t vertices remain. A fixed min-cut C of size k survives a single contraction step with probability $\geq 1 - 2/t$. By induction, survival probability $\geq \binom{t}{2} / \binom{n}{2}$. Setting $t = n/\sqrt{2}$ and recursing twice gives $O(n^2 \log n)$ expected time. The success probability is $\Omega(1/\log n)$, and repeating $O(\log^2 n)$ times gives high probability.

Solution 19:

Construct a diagonalizing TM D that on input x runs M_x on x for $f(|x|)$ steps and outputs the opposite answer. If D runs in $o(f(n)/\log f(n))$ time, we get a contradiction for all sufficiently large x . The $\log f(n)$ overhead comes from simulation costs: D must simulate M_x which has a different tape alphabet and number of tapes.

Solution 20:

Given set cover instance $(U, \mathcal{S}, k)$, construct a directed graph with source s , set vertices s_j , element vertices u_i , edges $s \rightarrow s_j$ (cost 1), $s_j \rightarrow u_i$ if $u_i \in S_j$ (cost 0). Steiner tree connecting s to all u_i uses $\leq k$ set vertices iff cover of size k exists. The approximation uses randomized rounding and the junction tree theorem.

Solution 21:

Progress: if $\vdash e : \tau$ and e is closed, then e is a value or $e \rightarrow e'$. Proof by induction on typing derivation. Preservation: if $\vdash e : \tau$ and $e \rightarrow e'$, then $\vdash e' : \tau$. Proof by induction on the reduction relation with substitution lemma. For System F, add type abstraction and application cases, with type substitution preserving the subject reduction property.

Solution 22:

For each variable x_i , create a chain of vertices $v_{i,1}, \dots, v_{i,k}$ with edges forming a path that can be traversed left-to-right (true) or right-to-left (false). For each clause $C_j = (\ell_1 \vee \ell_2 \vee \ell_3)$, insert a clause vertex connected to the corresponding literal vertices in

the variable chains. A Hamiltonian cycle exists iff we can pick a consistent direction for each variable chain that hits all clause vertices. This is the standard NP-completeness reduction.

Solution 23:

Nisan's pseudorandom generator for logspace: $G : \{0,1\}^{O(\log^2 n)} \rightarrow \{0,1\}^n$ fools any logspace machine. The construction uses the INW generator (impagliazzo-Nisan-Wigderson) based on extractors. By enumerating all $2^{O(\log^2 n)}$ seeds, we can simulate RL in SC (polynomial time, polylog space). This gives $RL \subseteq SC \subseteq P$.

Solution 24:

If $P = NP$, then $PH = P$ (polynomial hierarchy collapses). Since $BPP \subseteq \Sigma_2^P \cap \Pi_2^P$ (by Sipser-Gacs theorem), we get $BPP \subseteq P$. The inclusion $BPP \subseteq EXP$ follows from exhaustive simulation of the probabilistic computation over all random bits. Whether $P = BPP$ is open but believed true due to pseudorandom generators.

Solution 25:

Let $\mathcal{P}$ be a nontrivial property of partial recursive functions, i.e., there exist f, g with $f \in \mathcal{P}$, $g \notin \mathcal{P}$. Assume $\mathcal{P}$ is decidable. Reduce the halting problem to it: for input (M, x) , construct M' that on input y simulates M on x and then behaves like f (if $\emptyset \notin \mathcal{P}$) or like g (if $\emptyset \in \mathcal{P}$). Then M halts on x iff $M' \in \mathcal{P}$, a contradiction.